



5779 - A User's Guide to Lyman-alpha Damping Wings in Reionization-Epoch Galaxies

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST has recently discovered a trio of reionization-epoch galaxies whose stellar continua show clear evidence of Lyman-alpha damping wing absorption. Surprisingly, the absorption cannot originate in the intergalactic medium (IGM). Instead, it points to a substantially-neutral circumgalactic medium (CGM). This is a novel probe of both the timing and the topology of hydrogen reionization, and we will write its "User's Guide." CGM damping wings complement metal-ion tests including the MgII and OI forests because interpreting them does not depend on the unknown metallicity distribution. They complement Lyman-alpha emitter statistics because they do not reflect complicated resonant scattering, and they are sensitive primarily to the CGM rather than the IGM. Their discovery raises many questions: How do the reionization histories of the CGM and IGM relate? To what extent are CGM damping wings a barometer of the metagalactic ionizing background that drove reionization? To what extent are they sensitive to small-scale processes such as inefficient inflows or feedback from star formation? We propose a two-year study to address these and related questions. We will anchor our work in comparisons between JWST observations and detailed analysis of cosmological radiation hydrodynamic simulations. The simulations are known to match a variety of galaxy and reionization observables. Encouragingly, they also readily predict that the absorbing HI column varies strongly with halo mass and redshift as observed by JWST. We will publish our predictions in a way that

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encourages further observational tests, and our methods in a way that encourages complementary modeling efforts.

OBSERVING DESCRIPTION